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Megan S. Reich

EDUCATION

2019-2024 PhD Biology, Department of Biology, University of Ottawa, Canada

- Thesis topic: Advancements in isotopic geolocation tools for insect migration research
  - Nominated for best thesis of 2024 at the University of Ottawa
  - Advisors: Profs Clément Bataille and Heather Kharouba

2018-2019
- MSc Earth Sciences; fast-tracked to PhD

2011-2015 BSc Environmental Sciences, University of British Columbia, Canada (with distinction)

2013
- International exchange to the University of New South Wales, Australia

2008-2010
- 1st and 2nd year at the University of the Fraser Valley, Canada

CURRENT POSITION

2026-present Postdoctoral Research Associate, SAiVE lab with Prof Bataille, Department of Forestry and Natural Resources, Purdue University, USA

2024-2026 Postdoctoral Fellow, SAiVE lab with Prof Bataille*,* Department of Biology, University of Ottawa, Canada

RECOGNITION

2023 Mitacs Globalink Research Award
2022-2023 Ontario Graduate Scholarship
2021 Best graduate talk, 18th Annual Ottawa-Carleton Institute of Biology Symposium
2021-2022 Ontario Graduate Scholarship
2020-2023 Excellence Scholarship-Doctorate, University of Ottawa
2020 ORIGIN Graduate Fellowship 2021, University of Utah
2020-2021 Queen Elizabeth II Graduate Scholarship in Science and Technology
2020 Danks Scholarship, Entomological Society of Canada
2019-2020 Admission Scholarship-Doctorate, University of Ottawa
2018-2019 Admission Scholarship-Master's, University of Ottawa

PUBLICATIONS

Summary

I have contributed to 16 peer-reviewed articles, plus two articles currently in review. I am the first and corresponding author for four of these publications (Reich et al., 2021, 2023, 2024 and 2025). I am also senior author on the publication that resulted from an honours thesis project that I supervised (Lindroos et al., 2023). As of May 2026, I have 249 citations with an H-index of 11 and i10 index of 12 on Google Scholar.

Manuscripts in Review

18. Boman, J., Díaz, G., Vilaseca, Iu, **Reich, M. S.**, Backström, N., Talavera, G. (*in review*). An epigenetic aging clock in butterflies.
17. Brlík, V., Samaš, P., Barshep, Y., Ivande, S., Okposio, E. O., Trubač, J., Vondrovicová, L., Nwaogu, C., **Reich, M.**, Stricker, C., Mikula, P., Prochazka, P. (*in review*). Environment-isotope associations are consistent across temporal scales but vary across regions and bird species.

Peer-reviewed Publications

16. Ragyák., Á., Varga, T., Ahmed, N. A. K., Braun, M., Horváth, A., Lisztes-Szabó, Z., Szabó, S., Kaste, J., Bataille, C. B., **Reich, M.**, Baranyai, E., Sajtos, Z. (2026). Comprehensive analysis of the elemental composition and isotope ratios of honeys from US East Coast. *Food chemistry,* 515: 149254. doi: 10.1016/j.foodchem.2026.149254

15. Angell, C., **Reich, M.** (*accepted – in press*). Aggregation of *Boreopiophila tomentosa* (Diptera: Piophilidae) on discarded moose antlers. *The Journal of the Entomological Society of Ontario*.
14. Dargent, F., **Reich, M.S.**, Miller, M., Studens, K., Benvidi, N., Perrault, K., Aibueku, J., Holmes, B., Bataille, C., Candau, J-N. (*accepted - in press*). An integrated framework to identify and characterize regional-scale insect dispersal. *Ecological Applications*.
13. **Reich, M. S.**, Shipilina, D., Talla, V., Bahleman, F., Kébé, K., Berger, J. L., Backström, N., Talavera, G., Bataille, C. P. (2025). Isotope geolocation and population genomics in *Vanessa cardui*: Short- and long-distance migrants are genetically undifferentiated. *PNAS Nexus,* 4(2): pgae586. doi: 10.1093/pnasnexus/pgae586
12. Le Corre, M., Dargent, F., Grimes, V., Wright, J., Côté, S. D., **Reich, M. S.**, Candau, J-N., Miller, M., Holmes, B., Bataille, C. P., Britton, K. (2025). An ensemble machine learning bioavailable strontium isoscape for Eastern Canada. *FACETS*, 10: 1-17. doi: 10.1139/facets-2024-0180
11. **Reich, M. S.**, Ghouri, S., Zabudsky, S., Hu, L., Le Corre, M., Ng'iru, I., Benyamini, D., Shipilina, D., Collins, S. C., Martins, D. J., Vila, R., Talavera, G., Bataille, C. P. (2024) Trans-Saharan migratory patterns in Vanessa cardui and evidence for a southward leapfrog migration. *iScience*, 27, 111342. doi: 10.1016/j.isci.2024.111342
10. Suchan, T., Bataille, C. P., **Reich, M. S.**, Toro-Delgado, E., Vila, R., Pierce, N. E., Talavera, G. (2024). A trans-oceanic flight of over 4,200 km by painted lady butterflies. *Nature Communications,* 15(1), 5205. doi: 10.1038/s41467-024-49079-2
9. Gorki, J. L., López-Mañas, R., Sáez, L., Menchetti, M., Shapoval, N., Andersen, A., Benyamini, D., Daniels, S., García-Berro, A., **Reich, M. S.**, Scalercio, S., Toro-Delgado, E., Bataille, C. P., Domingo-Marimon, C., Vila, R., Suchan, T., Talavera, G. (2024). Pollen metabarcoding reveals the origin and multigenerational migratory pathway of an intercontinental-scale butterfly outbreak. *Current Biology.* S0960982224006808. doi: 10.1016/j.cub.2024.05.037
8. Ghouri, S., **Reich, M. S.**, Lopez-Mañas, R., Talavera, G., Bowen, G., Vila, R., Talla, V. N. K., Collins, S. C., Martins, D. J., & Bataille, C. (2024). A hydrogen isoscape for tracing the migration of herbivorous lepidopterans across the Afro-Palearctic range. *Rapid Communications in Mass Spectrometry*, 38(3), e9675. doi: 10.1002/rcm.9675
7. Talavera, G., García-Berro, A., Talla, V. N. K., Ng'iru, I., Bahleman, F., Kébé, K., Nzala, K. M., Plasencia, D., Marafi, M. A. J., Kassie, A., Goudégnon, E. O. A., Kiki, M., Benyamini, D., **Reich, M. S.**, López-Mañas, R., Benetello, F., Collins, S. C., Bataille, C. P., Pierce, N. E., Martins, D. J., Suchan, T., Menchetti, M., Vila, R. (2023). The Afrotropical breeding grounds of the Palearctic-African migratory painted lady butterflies (*Vanessa cardui*). *Proceedings of the National Academy of Sciences*, 120(16), e2218280120. doi: 10.1073/pnas.2218280120
6. Dargent, F., Candau, J.-N., Studens, K., Perrault, K. H., **Reich, M. S.**, & Bataille, C. P. (2023). Characterizing eastern spruce budworm's large-scale dispersal events through flight behavior and stable isotope analyses. *Frontiers in Ecology and Evolution*, 11, 1060982. doi: 10.3389/fevo.2023.1060982
5. **Reich, M.S.**, Kindra, M., Dargent, F., Hu, L., Flockhart, T., Norris, R., Kharouba, H., Talavera, G., Bataille, C.P. (2023) Metals and metal isotopes incorporation in insect wings: Implications for geolocation and pollution exposure. *Frontiers in Ecology and Evolution*, 11. doi: 10.3389/fevo.2023.1085903
4. Lindroos, E.E., Bataille, C.P., Holder, P.W., Talavera, G., **Reich, M.S** (2023). Temporal stability of δ^2H in insect tissues: Implications for isotope-based geographic assignments. *Frontiers in Ecology and Evolution*, 11. doi: 10.3389/fevo.2023.1060836
3. López-Mañas, R., Pascual-Díaz, J.P., García-Berro, A., Bahleman, F., **Reich, M.S.**, Pokorny, L., Bataille, C.B., Vila, R., Domingo-Marimon, C., Talavera, G. (2022). Erratic spatiotemporal vegetation growth anomalies drive population outbreaks in a trans-Saharan insect migrant. *Proceedings of the National Academy of Sciences*, 119:19. 3-5. doi: 10.1073/pnas.2121249119
2. **Reich, M.S.**, Flockhart, D.T.T., Norris, D.R., Hu, L., Bataille, C.P. (2021). Continuous-surface geographic assignment of migratory animals using strontium isotopes: A case study with monarch butterflies. *Methods in Ecology and Evolution,* 1-13. doi: 10.1111/2041-210X.13707.
1. Amundrud, S.L., Clay-Smith, S.A., Flynn, B.L., Higgins, K.E., **Reich, M.S.**, Wiens, D.R.H., & Srivastava, D.S. (2019). Drought alters the trophic role of an opportunistic generalist in an aquatic ecosystem. *Oecologia*, 189:3. 1–12. doi: 10.1007/s00442-019-04343-x

RESEARCH FUNDING

Awarded Funding

2026-2030 Australian Research Council, Discovery Project
"Exchange networks and social resilience across the last deglaciation."
Chief Investigators: A. Mackay and A. Dosseto; **Role: Partner investigator (.1 FTE)**
Funding: AUD$741,226

2024-2025 Percy Sladen Memorial Fund Grant. The Linnean Society. (United Kingdom)
"Escaping the heat: Unravelling the migratory connectivity of aestivating Bogong Moths *Agrotis infusa* in the Australian Alps through a multi-isotope framework."
Principal Investigator: Megan Reich
Funding: £2,000

2024 BayCenSI Stepping Stones funding. Bayreuth Center for Stable Isotope Research in Ecology and Biogeochemistry (Germany)
"Exploring the migratory patterns of the pioneer caper white (*Belenois aurota*) and the African migrant (*Catopsilia florella*) with stable isotopes."
Principal Investigator: Megan Reich
Funding: in-kind (isotopic analysis)

2024-2027 Ministerio de Ciencia, Innovación y Universidades, Proyectos de I+D+i (PID2023-152239NB-I00) (Spain)
"Genomic and epigenomic signatures of migration in butterflies (MIGRASPHERE)."
Principal Investigator: G. Talavera; **Role: working team member**
Funding: 242,500€

2025-2028 Red de Parques Nacionales, Ministerio para la transición ecological y el reto demográfico (Spain)
"Seasonal migratory insect biodiversity in protected coastal areas: monitoring, connectivity and impact".
Principal Investigator: G. Talavera; **Role: working team member**
Funding: 117,600€

2023-2026 Natural Resources Canada, E.I.S. Small Research Fund (Canada)
"Tracking insect outbreak expansion: using moth trapping, genomics, and stable isotopic analyses to validate and refine understanding and predictions of regional moth dispersal."
Principal Investigator: C. Bataille; **Role: Postdoctoral researcher**
Funding: C$580,000

2022-2024 CSIC (LINKA20399) (Spain)
"An interdisciplinary scheme to advance in the field of ecology and evolution of insect migration."
Principal Investigator: G. Talavera; **Role: working team member**
Funding: 24,000€

2021-2024 Ministerio de Ciencia e Innovación, Proyectos de I+D+i (PID2020-117739GA-I00) (Spain)
"Behavioral and ecological genomics of insect migration (ENTOMIGROME)."
Principal Investigator: G. Talavera
Role: working team member
Funding: 196000€

TEACHING

Courses Taught

Summer 2025 Part-time Professor, University of Ottawa: Introduction to Environmental Sciences (EVS1101)
- Delivered engaging online lectures for 143 students, incorporating online tools (e.g., Wooclap)
- Designed the class syllabus, quizzes, and examinations
- Addressed academic misconduct following university protocols and reflected on how to best address the use of generative AI in the classroom

Guest Lectures

Spring 2026 Guest lecturer, Purdue University: Isotopes in Natural Resources (FNR59800)
- Lecture title: "Lead the Way: Applications of Lead Isotopes"

Fall 2025 Guest lecturer, Earlham College, Indiana, USA: Entomology (BIOL 362)
- Delivered a 60-minute lecture entitled, "Insect Migration," to 2nd and 3rd year BSc students

Fall 2024 Guest lecturer, University of Ottawa: Isotope Mapping and Provenance Applications ((GEO 5144)
- Delivered a 60-minute lecture entitled, "Geolocation and source tracing with lead isotopes," to 4th year BSc and graduate students

Teaching Assistantships

Fall 2022 University of Ottawa: Global Biogeochemical Cycles (GEO 5191)
- Provided student feedback on laboratory reports on the global carbon cycle based on box modelling with Insight Maker

Spring 2022 University of Ottawa: Entomology (BIO3333)
- Led laboratory sessions, guiding students through identifying insect families using dichotomous keys, behavioural experiments with bean beetles and Manduca sexta, and a cockroach dissection

Spring 2019 University of Ottawa: Introduction to Earth Systems (GEO1111)
- Invigilated and marked exams; held office hours and answered student's questions about Earth history, plate tectonics, and rock formations

Fall 2018 University of Ottawa: Environmental Issues (EVS3101)
- Guided students through literature reviews and provided feedback on presentations and written reports of self-directed group projects on environmental issues, such as coral bleaching and microplastics

STUDENT SUPERVISION and MENTORING

(* indicates students who are coauthors on published papers)

2024-present Research advising: PhD student J. Stewart (Queen Mary University of London)
2024 Member of Review Panel for Confirmation of Candidature: R. Lownds (Western Sydney University)
2023-2025 Mentorship of high school science project: E. Chiariello (Fox Lane High School)
2021-2022 Supervision of undergraduate thesis: E. Lindroos* (University of Ottawa)
2020-2021 Mentorship of undergraduate thesis: S. Ghouri* (University of Ottawa)
2019-2020 Mentorship of undergraduate thesis: M. Kindra* (University of Ottawa)

RESEARCH PRESENTATIONS

Conference Presentations (* presenter)

Reich, M.S.*, Talavera, G., Backström, N., Bataille, C.P. (May 2025). Assessing the Effects of Lead Exposure on the Flight Ability of Painted Lady Butterflies. Ontario Ecology Ethology Evolution Colloquium, Ottawa, Canada. [12 minute talk]

Reich, M.S.*, Ghouri, S., Talavera, G., Bataille, C. P. (July 2023). Isotope-based geographic assignment provides valuable insights into long-distance butterfly migration. Biology of Butterflies 2023, Prague, Czech Republic.

Talavera, G.*, Gorki, L., Toro-Delgado, E., López-Mañas, R., **Reich, M. S.**, Pascual-Díaz, J. P., García-Berro, A., Menchetti, M., Domingo-Marimon, C., Pierce, N. E., Vila, R., Suchan, T., Bataille, C. P. (July 2023). Migratory ecology and population dynamics of the Painted Lady butterfly, *Vanessa cardui*. Biology of Butterflies 2023, Prague, Czech Republic.

Ghouri, S.*, Talavera, G., **Reich, M. S.**, Bataille, C. P. (July 2023). Hydrogen and strontium isoscapes for the African Palearctic range to reconstruct insect migration and connectivity. Biology of Butterflies 2023, Prague, Czech Republic.

Bataille, C.P.*, **Reich, M.S.,** Hassler, A. (July 2023). Metals and their isotopes: An opportunity to study insect ecology and physiology. Goldschmidt 2023, Lyon, France.

Reich, M.S.*, Ghouri, S., Zabudsky, S., Talavera, G., Bataille, C.P. (April 2023). There and back again: Combining hydrogen and strontium isotopes refines the trans-Saharan migratory patterns of the butterfly *Vanessa cardui*. European Geosciences Union (EGU) General Assembly 2023, Vienna, Austria.

Talavera, G.*, Gorki, L., Toro-Delgado, E., López-Mañas, R., **Reich, M.,** Menchetti, M., Domingo-Marimon, C., Sáez, L., Pierce, N., Vila, R., Bataille, C., Suchan, T. (April 2023). Migration ecology in insects: integrative approaches to trace long-distance movements of the Painted Lady butterfly (*Vanessa cardui*). European Geosciences Union (EGU) General Assembly 2023, Vienna, Austria.

Reich, M.S.*, Shipilina, D., Talla, V., Bahleman, F., Khebe, K., Talavera, G., Bataille, C.P., Backström, N. (Nov 2022). Lack of population structure between trans-Saharan migrants for the butterfly *Vanessa cardui* revealed by hydrogen and strontium isotope-based geographic assignment and genomics. 2022 ESA, ESC, and ESBC Joint Annual Meeting, Vancouver, Canada. [20 minute talk]

Dargent, F.*, Benvidi, N., Candau, J-N., **Reich, M.S.,** Bataille, C.P. (Nov 2022). Dual sulfur-hydrogen assignment of a boreal pest species (*Choristoneura fumiferana*) using a novel foliar sulfur isoscape for eastern Canada. 2022 ESA, ESC, and ESBC Joint Annual Meeting, Vancouver, Canada.

Reich, M.S.*, Lindroos, E., Kindra, M.K., Dargent, F., Hu, L., Flockhart, D. T. T., Norris, D.R., Talavera, G., Kharouba, H.M., Bataille, C.P. (September 2022). Testing the assumptions of geolocation using metals and metal isotopes in insect wings. 159th Annual General Meeting of the Entomological Society of Ontario, Virtual Meeting. [12 minute talk]

Reich, M.S.*, Lindroos, E., Kindra, M.K., Dargent, F., Hu, L., Kharouba, H.M., Bataille, C.P. (June 2022). Are insect wings really 'inert'? Testing a core assumption of isotope-based geographic assignment. IsoEcol: 12th International Conference on the Applications of Stable Isotope Techniques to Ecological Studies, 2022, Gaming, Austria. [20 minute talk]

Reich, M.*, Flockhart, D. T. T., Norris, D. R., Hu, L., Bataille, C.P. (June 2021). Geographic assignment of monarch butterflies using strontium isotopes. 18th Annual Ottawa-Carleton Institute of Biology Symposium, 2021, Virtual Meeting. [12 minute talk]

Reich, M.*, Flockhart, D. T. T., Norris, D. R., Bataille, C.P. (Nov 2020). Combining strontium and hydrogen isotopes to estimate the provenance of monarch butterflies. Entomological Society of America Annual Meeting, 2020, Virtual Meeting. [10 minute talk]

Poster Presentations (own presentations only)

Reich, M.S., Ghouri, S., López-Mañas, R., Suchan, T., Talavera, G., Bataille, C.P. (Jul 2024). Isotopic insights into the migratory patterns of the painted lady butterfly *Vanessa cardui* during an outbreak year. IsoEcol: 13th International Conference on the Applications of Stable Isotope Techniques to Ecological Studies, Fredericton, Canada. [Poster]

Reich, M., Flockhart, D. T. T., Norris, D. R., Hu, L., Bataille, C.P. (May 2021). Continuous-surface geographic assignment of monarch butterflies using strontium isotopes. IsoEcol: 11.5 International Conference on the Applications of Stable Isotope Techniques to Ecological Studies, 2021, Virtual Meeting. [Poster]

Reich, M., Flockhart, D. T. T., Norris, D. R., Bataille, C.P. (Aug 2020). Conservation of monarch butterflies using a novel strontium isotope geolocation tool. Ecological Society of America Annual Meeting, 2020, Virtual Meeting. [Poster]

Invited Presentations (own presentations only)

Reich, M.S. (Feb 2024). Multi-isotope geographic assignment for tracing the migratory connectivity of monarch butterflies. Trilateral Scientific Group Meeting on the Monarch Butterfly, Mexico City, Mexico: oral presentation

Reich, M.S. (May 2022). Isotope tools for geolocation of migratory insects. 1st Meeting on Butterfly Migration, Barcelona, Spain: oral presentation

Reich, M.S. (Feb 2021). Geographic assignment of migratory butterflies using hydrogen and strontium isotopes. International Day of Women and Girls in Science Seminar, Department of Earth and Environmental Sciences, University of Ottawa, Canada: oral presentation

CONSERVATION and POLICY

Feb 2025 Bogong Moth Summit, Canberra, Australia
Invited as part of the Bogong Watch Project Team to advise on isotope-based methods to map the Bogong moth migration with the intent of informing conservation activities for the recovery of the endangered population

Feb 2024 Trilateral Scientific Group Meeting on the Monarch Butterfly, Mexico City, Mexico
Invited as part of a Canadian delegation to Mexico to develop a science-based action plan for the recovery of the monarch butterfly, based on my expertise with respect to migration ecology

PROFESSIONAL SERVICE

Committees and Associations

2020-2022	Biology Graduate Student Association – Executive Member Councillor
2020-2022	Graduate Students Association des Étudiant.e.s Diplômé.e.s (GSAÉD)–Director
2020-2021	Informal Seminar Without A Name (ISWAN) – Organizer
2020-2021	GSAÉD Social Committee – Member
2011-2014	UBC Environmental Sciences Student Association – Executive Member

Conference organization

2022 ESA, ESC, and ESBC Joint Annual Meeting, Vancouver, BC, Canada
- Organized a member symposium: "Leveraging Isotopic Tools to Understand Insect Ecology"

Grant Peer-reviewing

2025 Deutsche Forschungsgemeinschaft (German Research Foundation)

Manuscript Peer-reviewing

Isotopes in Environmental & Health Studies, Avian Conservation and Ecology, Russian Entomological Journal, iScience, Scientific Reports, Proceedings of the Royal Society B, Ecography, Biological Journal of the Linnean Society, Science of the Total Environment, Micron

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PROFESSIONAL EXPERIENCE

Academic or Research Stays

Feb 2025 Western Sydney University, Australia
- Visited the lab of Prof Kate Umbers with an honorary academic appointment as a Visiting Fellow with the School of Science to collaborate on the ongoing project, "Escaping the heat: Unravelling the migratory connectivity of aestivating Bogong Moths Agrotis infusa in the Australian Alps through a multi-isotope framework"
- Fieldwork funded by the Percy Sladen Memorial Fund Grant from The Linnean Society

Jan-Jul 2023 Uppsala University, Sweden, and the Institut Botànic de Barcelona (CSIC), Spain
- Research stay during PhD to collaborate with Profs Niclas Backström and Gerard Talavera on an ongoing non-thesis research project, "Effects of host plant quality on the transcriptional signatures of flight in migratory butterflies"
- Visit funded by a Mitacs Globalink Research Award

Oct-Nov 2019 Institut Botànic de Barcelona (CSIC), Spain
- Visited IBB to launch PhD research projects in collaboration with Gerard Talavera

Non-academic Work Experience

Jan 2021-Aug 2022 Greenhouse staff, University of Ottawa
- Maintenance and monitoring of plants in the Department of Biology's research greenhouse

May 2019-Aug 2019 Assistant, Canadian Museum of Nature - Research Collection
- Museum curation in the entomological collection (beetles): specimen preparation, organization, and labelling

Nov 2017-Mar 2018 Water Literacy Coordinator, Fraser Riverkeeper/Swim Drink Fish Canada
- Organized the 11th Annual Fraser River Clean-up, where 650 volunteers collected 12 tonnes of garbage from the riverbank
- Delivered water literacy presentations to schools and community groups to encourage sustainable use of our local waterbodies

May 2017-Sept 2017 Research Assistant, PhytoInformatix
- Performed pesticide screening trials and efficacy assessments in an agricultural setting

Mar 2016-Feb 2017 Field Coordinator in Nepal, Projects Abroad - Himalayan Mountain Conservation Project
- Analyzed survey data and wrote monthly reports to the Annapurna Conservation Area Project; responsible for data management
- Led international volunteers in biological surveys of butterflies, birds, primates, mammals, and herpetological species using non-invasive methods (visual surveys and camera traps)
- Conducted volunteer feedback interviews, reported incidents, and created action plans

May 2015-Aug 2015 British Columbia Tree Fruits Cooperative, Survey Technician
- Evaluated 110,000 cherries for insect infestation using a dissecting microscope in order to fulfill the Government of Japan's survey requirements

Aug 2015-Dec 2015	Field Assistant, University of British Columbia • Assisted PhD student in sampling aquatic insect ecosystems within bromeliads along an elevation gradient in Monteverde, Costa Rica to explore the potential effects of climate change on this system, resulting in my first publication
Sept 2014-Apr 2015	Government of Canada - Internal Integrity and Security, Marketing & Communications Strategy Agent • Facilitated Security Awareness Week to increase security awareness of employees • Organized an in-person focus session of managers and team leaders that reviewed policies and procedures to generate constructive feedback and recommendations, and create new user guides

Volunteering

2018-2020	Canadian Museum of Nature - Entomological Research Collection • Mounted and labelled specimens, organized the collection, and performed other curatorial tasks
2011-2015; 2018-2021	Girl Guides of Canada – Unit Leader for the 16th Guides Ottawa and 43rd Guides Vancouver • Mentored girls aged 9-11 in games, activities, and community involvement to develop their leadership skills • Organized safe camping trips; Managed unit finances

MEDIA and OUTREACH

Science Journalism

Reich, M. (2021). "Strontium isotopes can map monarch butterfly migrations and help conservation efforts". The Conversation (Canada), September 28, 2021. Available at: https://theconversation.com/strontium-isotopes-can-map-monarch-butterfly-migrations-and-help-conservation-efforts-168031?utm_source=dlvr.it&utm_medium=twitter

Selected Media Coverage

Sheena Goodyear (Apr 2025). "Chasing butterflies around the globe changed this photographer's worldview." CBC Radio: As It Happens. https://www.cbc.ca/radio/asithappens/painted-lady-butterfly-migration-1.7515878

Jesse Greenspan (Apr 2025). "This Butterfly's Epic Migration Is Written into Its Chemistry." Scientific American. https://www.scientificamerican.com/article/this-butterflys-epic-migration-is-written-into-its-chemistry/

Alan Neal (Feb 2025). "The Odyssey of the Painted Lady Butterfly." CBC's All in a Day with Alan Neal, CBC Radio. https://www.cbc.ca/listen/live-radio/1-92-all-in-a-day/clip/16128076-the-odyssey-painted-lady-butterfly

Diego Sánchez Martínez (Aug 2024). "El misterio de las mariposas que aparecieron al otro lado del Atlántico." El País. https://elpais.com/clima-y-medio-ambiente/2024-08-26/el-misterio-de-las-mariposas-que-aparecieron-al-otro-lado-del-atlantico.html#?prm=copy_link

Andrew Carter. (July 2022). "The Andrew Carter Morning Show (Monday July 25, 2022)." The Andrew Carter Podcast, CJAD 800AM, iHeart Radio. https://omny.fm/shows/the-andrew-carter-morning-show/the-andrew-carter-morning-show-monday-july-25-2022

Nicole Chu. (Oct 2021). "Tracking isotopic "fingerprints" of monarch butterflies." Episode 92, Beats Research Radio, University of Ottawa Heart Institute. http://beatsresearch.com/Radio.php

David Frey. (Oct 2021). "Isotope mapping sheds light onto monarch journeys." The Wildlife Society. https://wildlife.org/isotope-mapping-sheds-light-onto-monarch-journeys/

Maryam Rana. (Sep 2021). "Monarch Butterflies: From strontium isotope mapping to migratory routes." The Fulcrum. https://thefulcrum.ca/sciencetech/monarch-butterflies-from-strontium-isotope-mapping-to-migratory-routes/

Classroom Outreach

Nov 2025	Blackwood Elementary, Grade 4 [via Skype a Scientist]
Oct 2025	Lisha Kill Middle School, Grade [via Skype a Scientist]
Sep 2025	Knudson Academy of the Arts, Grade 6 – 8 [via Skype a Scientist]
Jun 2022, May 2023	Guest speaker at Lac La Hache Elementary School, Grade 1-2
Feb 2021, Apr 2021	Guest speaker at Scriber Lake High School
Sep 2022	University of Houston, Science Teaching [via Skype a Scientist]
Sep 2022	University of Houston, Effective Teaching Strategies: Science Education [via Skype a Scientist]
Oct 2021	Amana Academy, Grade 3 [via Skype a Scientist]

PROFESSIONAL SKILLS

Data analysis skills

- Expertise in spatial ecology and spatial modelling
- Expert in developing maps of isotopic variation ("isoscapes")
- Proficient with machine learning (e.g., random forest and ensemble machine learning)
- Experience with Bioinformatic Pipelines (whole-genome sequencing, RNA-seq)
- Experience with geometric morphometric analysis
- Confident with basic statistics (e.g., linear mixed models, PCA)

Computing skills

- Expertise in data analysis and visualization with R
- Experience with ArcGIS
- Experience with bash scripting and High-performance Computing
- Professional development workshops:

2024	Bioinformatics: Analysis of RNA-Sequencing Data, Compute Ontario
2024	Introduction to High-Performance Computing, IT Solutions, University of Ottawa
2020	Introduction to the UNIX/LINUX Commandline, Lund University
2020	Python Workshop, IT Solutions, University of Ottawa
2020	Programming in R Workshop, IT Solutions, University of Ottawa

Laboratory skills: Isotope Mass Spectrometry

- Extensive experience in geochemistry clean labs (UOttawa, Carleton University, UBC)
- Column chromatography (Sr, Pb, and Ca isotopes)
- Preparing organic samples for metal isotope analysis (Sr, Pb, Ca) and stable isotope analysis (S, H)
- Operating Multicollector Inductively Coupled Plasma Mass Spectrometers (MC-ICP-MS)
- Operating 8900 Triple Quadruple ICP-MS for trace element analysis
- Data quality control and analysis
- Professional development workshops:

2019	Spill Response Training, University of Ottawa
2018	WHMIS 2015 - for laboratory workers

Laboratory skills: Behavioural Ecology

- Experience with rearing insects and butterfly husbandry
- Designing and executing behavioral assays (e.g., Flight mill assays of migratory capacity)

Field Ecology

- Professional development:

exp: Nov 2027	Remote First Aid & CPR/AED Level C, OFA Level 1
2017	Field Technician, Canadian Aquatic Biomonitoring Network (CABIN)

- Extensive international field experience and competence:
    - 2 months in the Australian Alps collecting moth samples
    - Multiple trips to Europe collecting butterfly samples (e.g., Portugal, Spain, Cyprus, Malta)
    - 2 weeks fieldwork in Morocco
    - 2.5 months in the USA collecting plant samples
    - 12 months in the Annapurna region, Nepal as an ecotourism field coordinator
    - 3.5 months in Monteverde, Costa Rica as a research assistant

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LANGUAGE

English (native language)

PROFESSIONAL SOCIETIES

2026- present	Association for Women in Science
2024- present	Moths and Butterflies Australasia
2022-present	British Ecological Society
2022-present	Earth Science Women's Network
2015-19, 22, 25-present	Entomological Society of British Columbia
2018-2024	Entomological Society of Ontario
2015-2022	Entomological Society of Canada
2020	Ecological Society of America
2016-18, 2020	Entomological Society of America